SDITO Lab PC-CS6G4

Name : Swati Sahoo **Roll No** : CSE/23/085 **Exam Roll No** : 11600223131

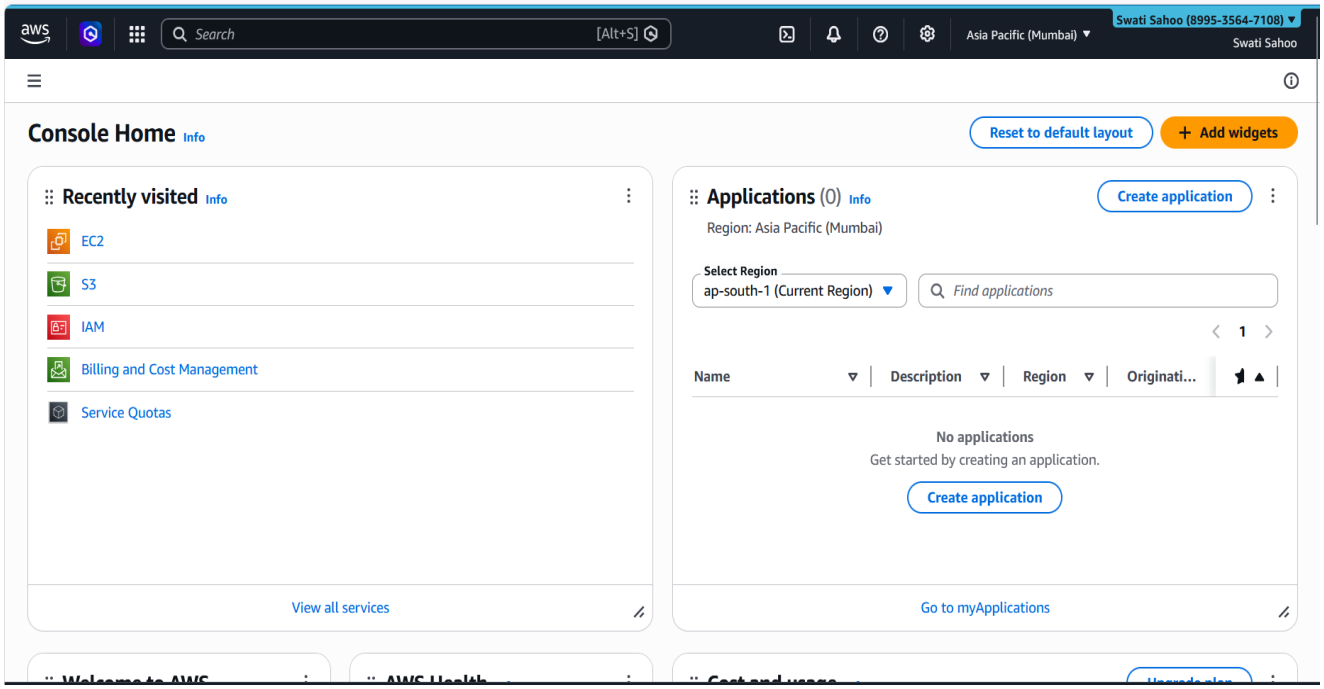
Assignment 12

Problem Statement : Deploy and Run the project in AWS without using the Port.

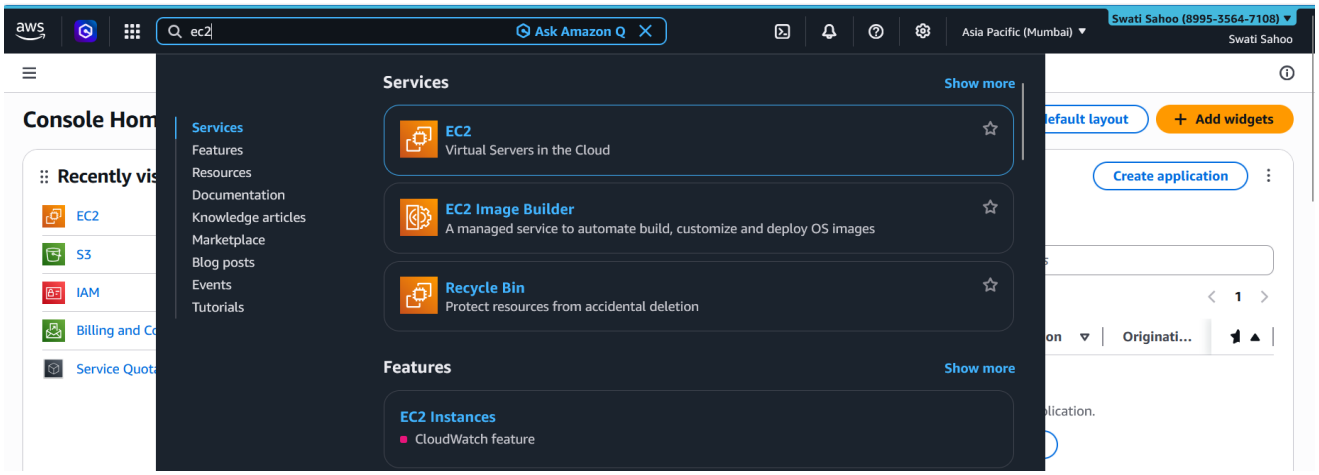
Solution:

STEP 1: Create a New Security Group(Skip this step if security group has already been created)

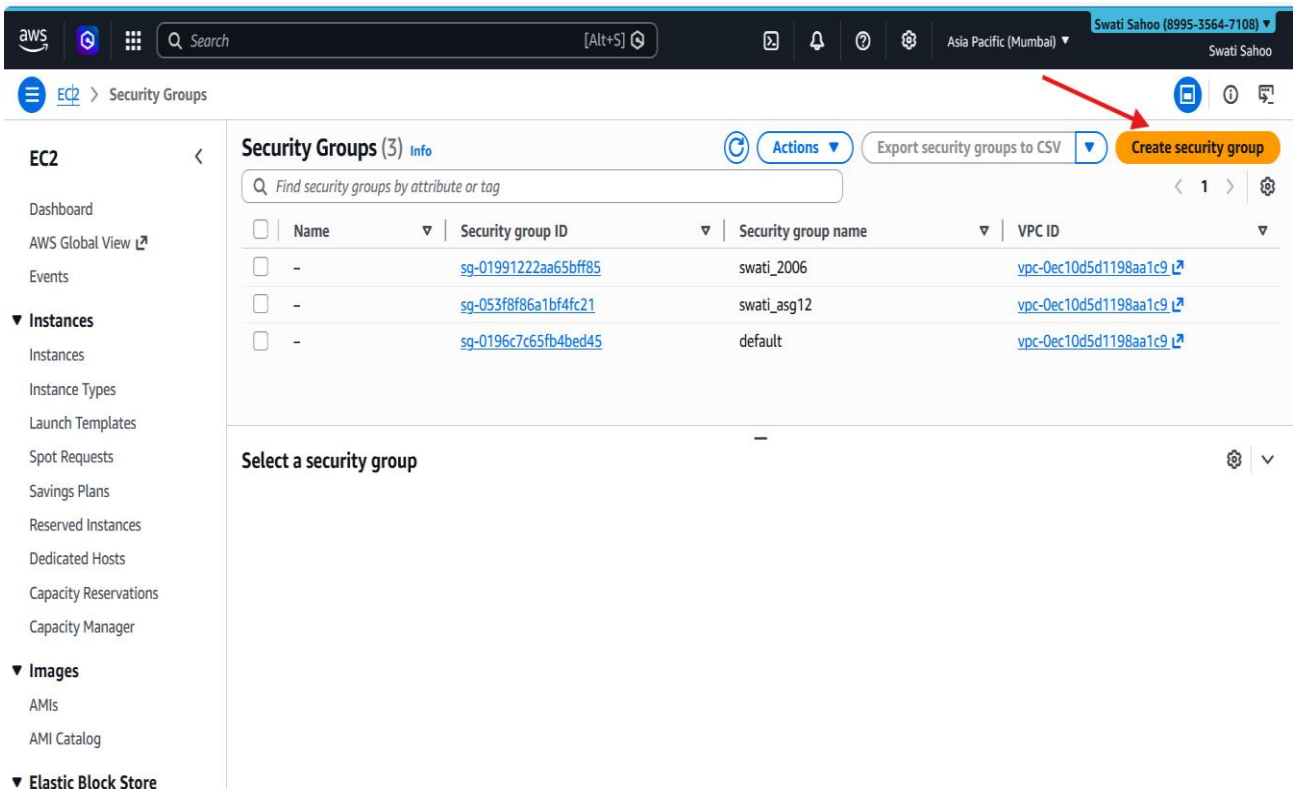
1. Login to AWS Console



2. **Go to EC2 Dashboard.**



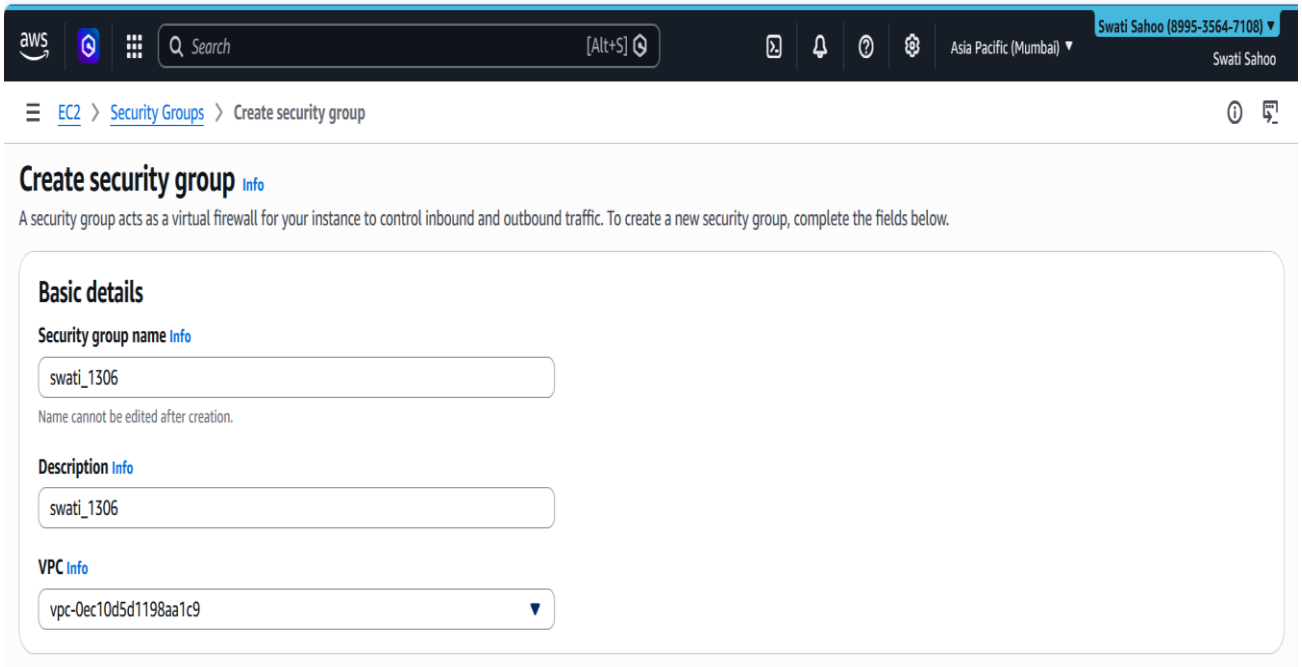
3. **Click Security Groups.**
4. Click Create Security Group



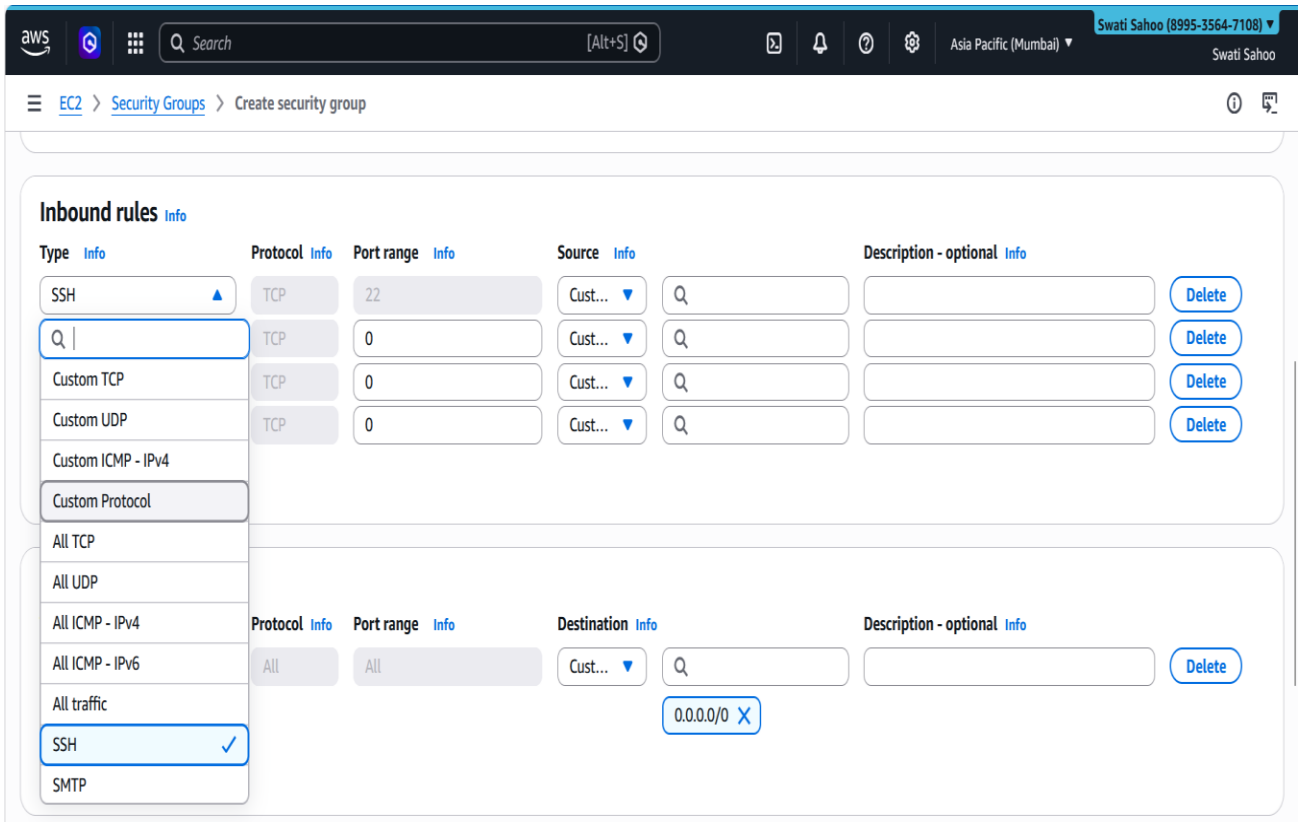
5.Enter:

- Security group name: swati_1306
- Description: swati_1306

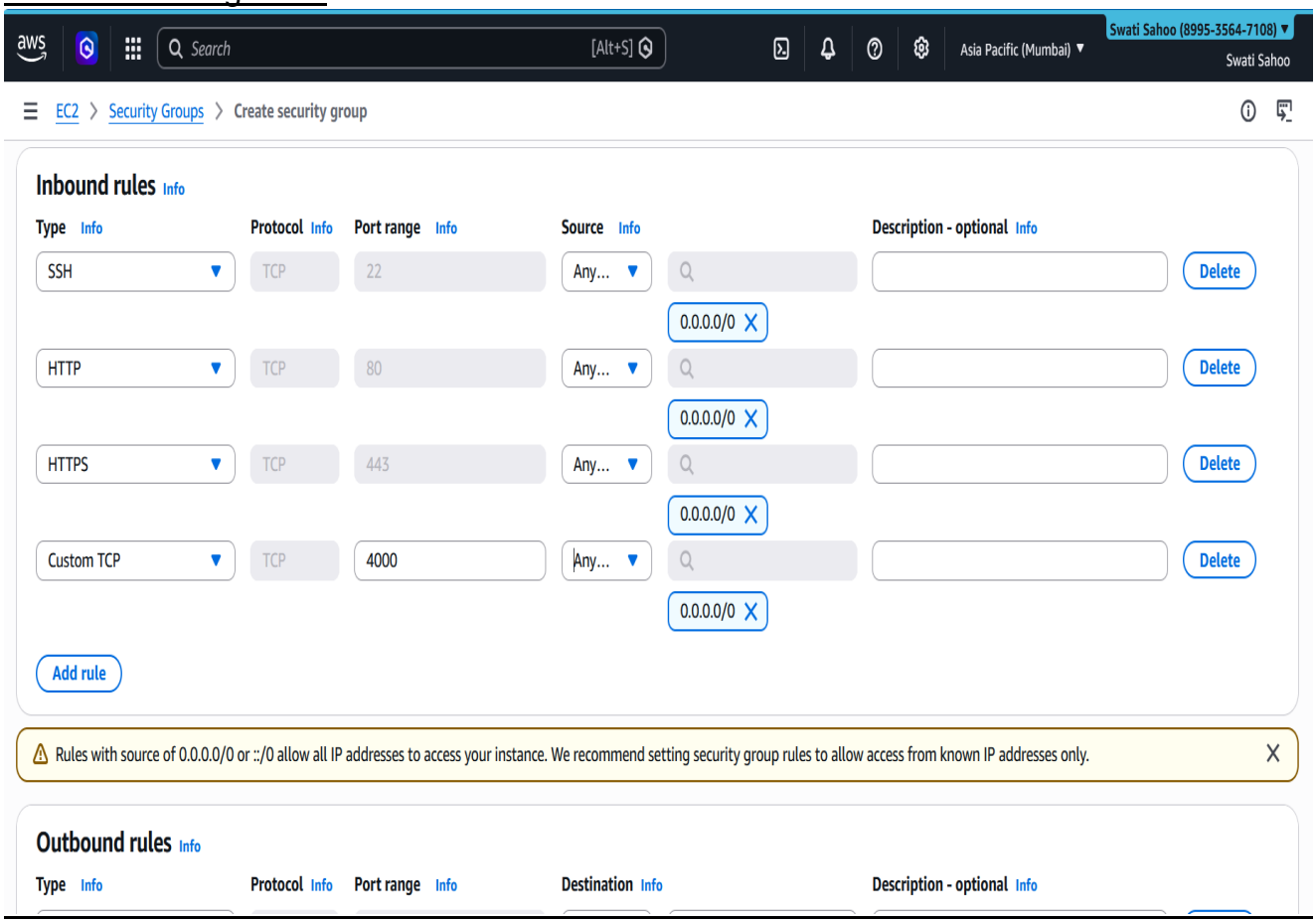
- Select default VPC



Add Inbound Rules



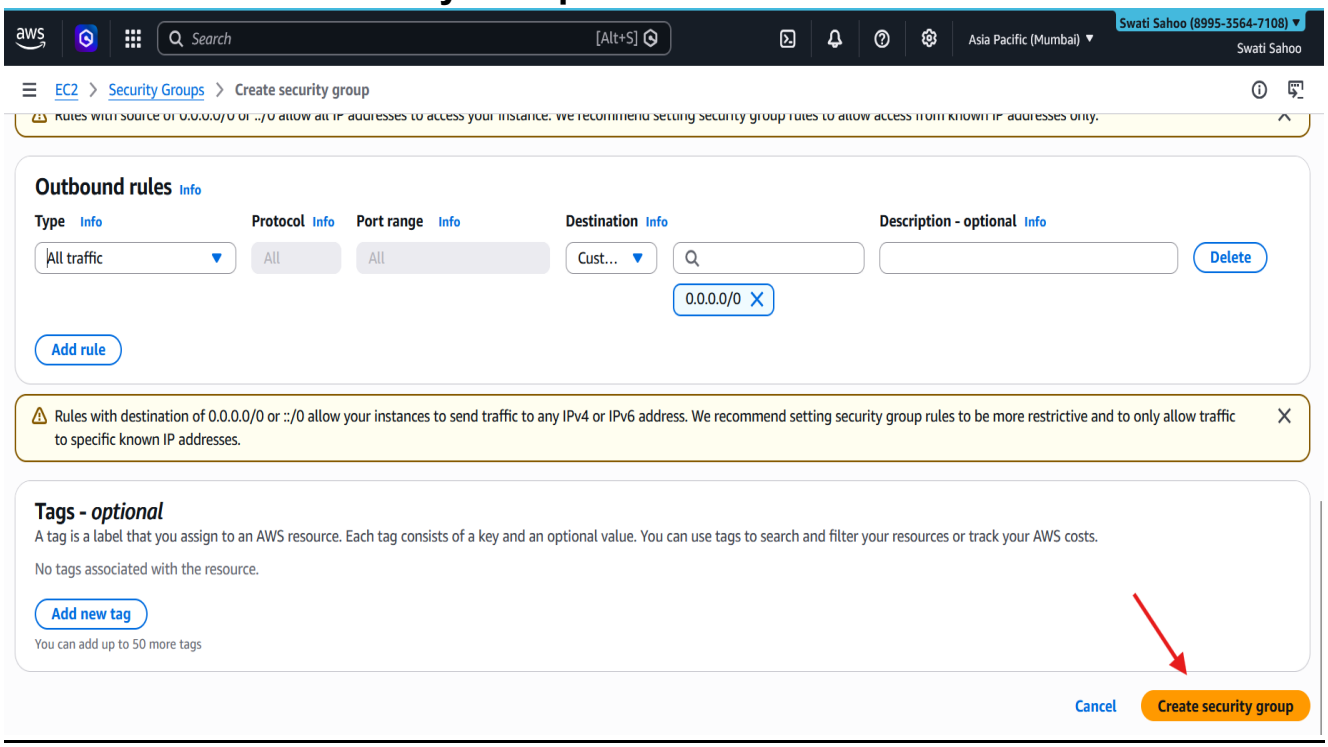
Add the following rules:



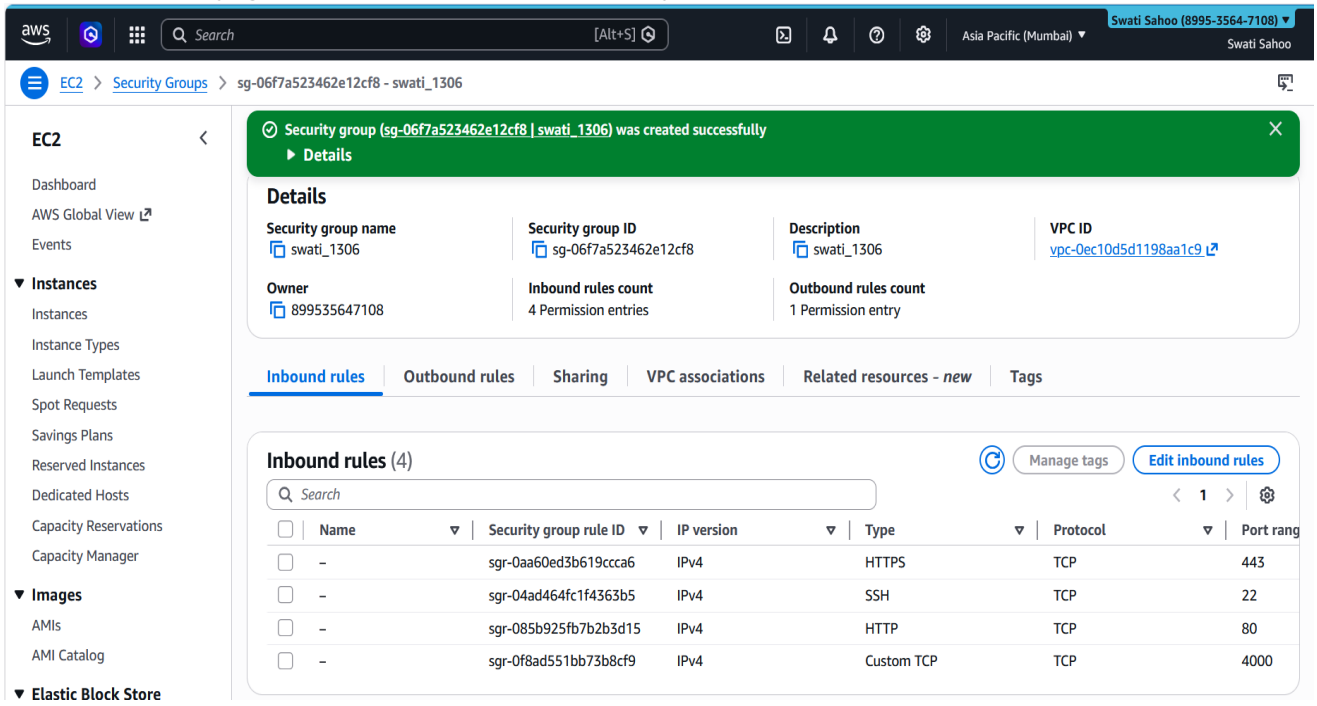
Why 4000?
Because I want to run my Node.js app on port 4000.

Type	Port	Source
SSH	22	0.0.0.0/0
HTTP	80	0.0.0.0/0
HTTPS	443	0.0.0.0/0
Custom TCP	4000	0.0.0.0/0

- Click **Create Security Group**

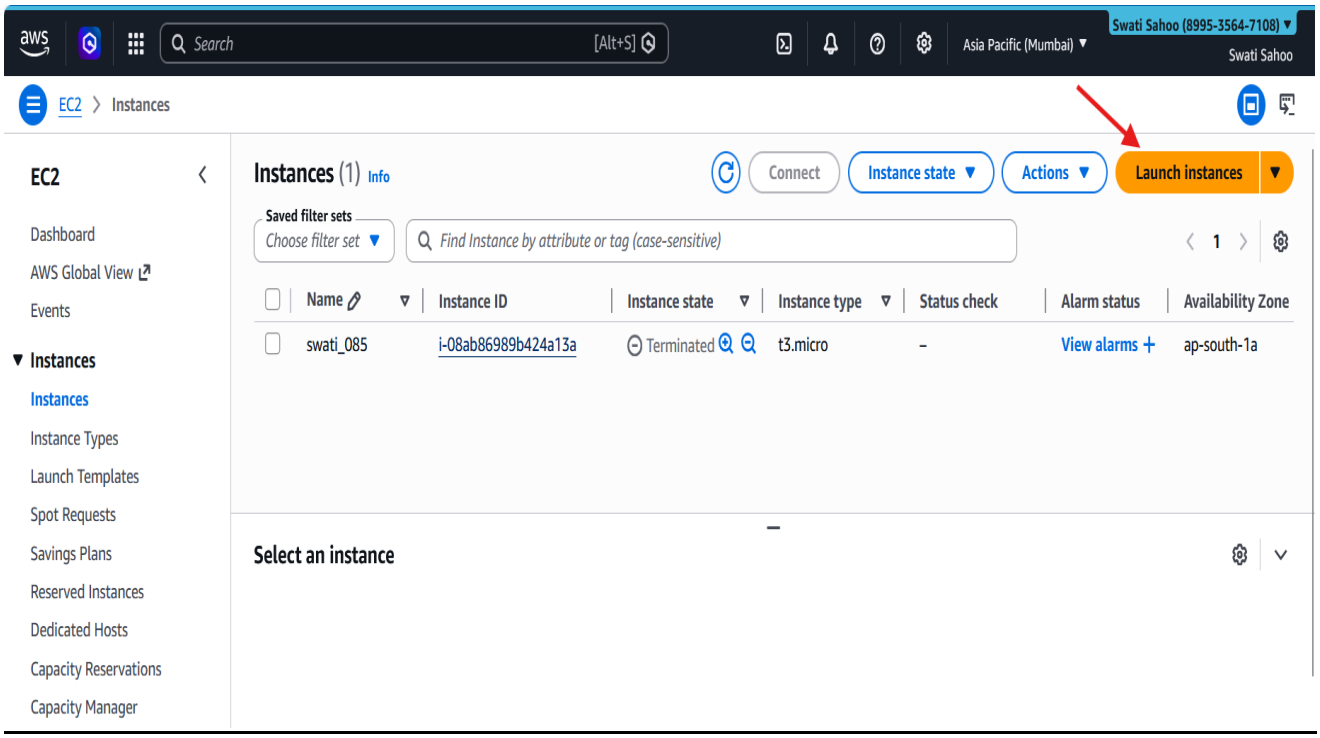


- Security group created successfully.



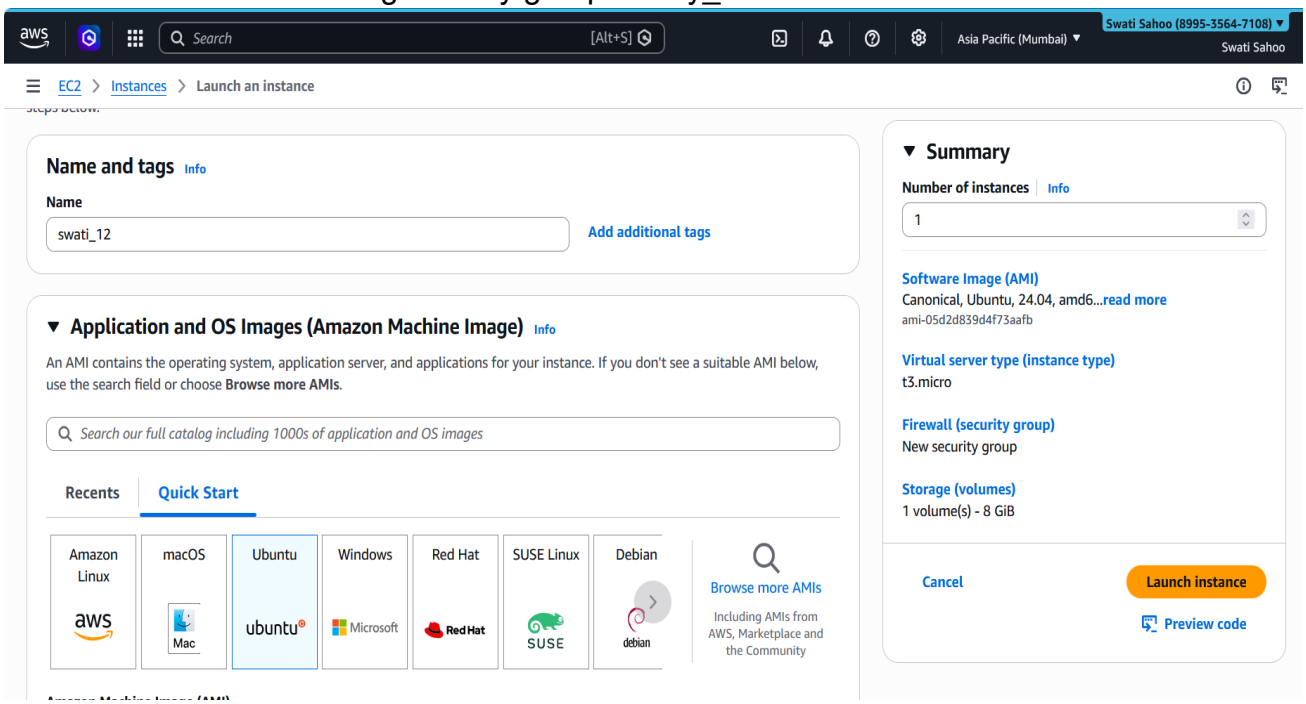
STEP 2: Launch EC2 Instance

1. Go to **EC2 → Instances**
2. Click **Launch Instance**



Configuration:

- **Name:** swati_12
- **AMI:** Ubuntu 24.04
- **Instance Type:** t3.micro
- **Key pair:** Select your existing key pair
- **Network Settings:**
  - Enable Auto-assign Public IP
  - Select existing security group → key_13



aws | Search [Alt+S] | Asia Pacific (Mumbai) | Swati Sahoo (8995-3564-7108) | Swati Sahoo

EC2 > Instances > Launch an instance

Instance type Info | Get advice

Instance type

t3.micro — Free tier eligible

Family: t3 2 vCPU 1 GiB Memory Current generation: true
On-Demand Linux base pricing: 0.0112 USD per Hour
On-Demand SUSE base pricing: 0.0112 USD per Hour
On-Demand Windows base pricing: 0.0204 USD per Hour
On-Demand Ubuntu Pro base pricing: 0.0147 USD per Hour
On-Demand RHEL base pricing: 0.04 USD per Hour

All generations

Compare instance types

Additional costs apply for AMIs with pre-installed software

Key pair (login) Info

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - *required*

key_13 | Create new key pair

Summary

Number of instances Info

1

Software Image (AMI)
Canonical, Ubuntu, 24.04, amd6...read more
ami-05d2d839d4f73aafb

Virtual server type (instance type)
t3.micro

Firewall (security group)
-

Storage (volumes)
1 volume(s) - 8 GiB

Cancel | Launch instance

Preview code

aws | Search [Alt+S] | Asia Pacific (Mumbai) | Swati Sahoo (8995-3564-7108) | Swati Sahoo

EC2 > Instances > Launch an instance

Network settings Info

Edit

Network Info

vpc-0ec10d5d1198aa1c9

Subnet Info

No preference (Default subnet in any availability zone)

Auto-assign public IP Info

Enable

Firewall (security groups) Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

Create security group | Select existing security group

Common security groups Info

Select security groups | Compare security group rules

swati_1306 sg-06f7a523462e12cf8 X
VPC: vpc-0ec10d5d1198aa1c9

Security groups that you add or remove here will be added to or removed from all your network interfaces.

Summary

Number of instances Info

1

Software Image (AMI)
Canonical, Ubuntu, 24.04, amd6...read more
ami-05d2d839d4f73aafb

Virtual server type (instance type)
t3.micro

Firewall (security group)
swati_1306

Storage (volumes)
1 volume(s) - 8 GiB

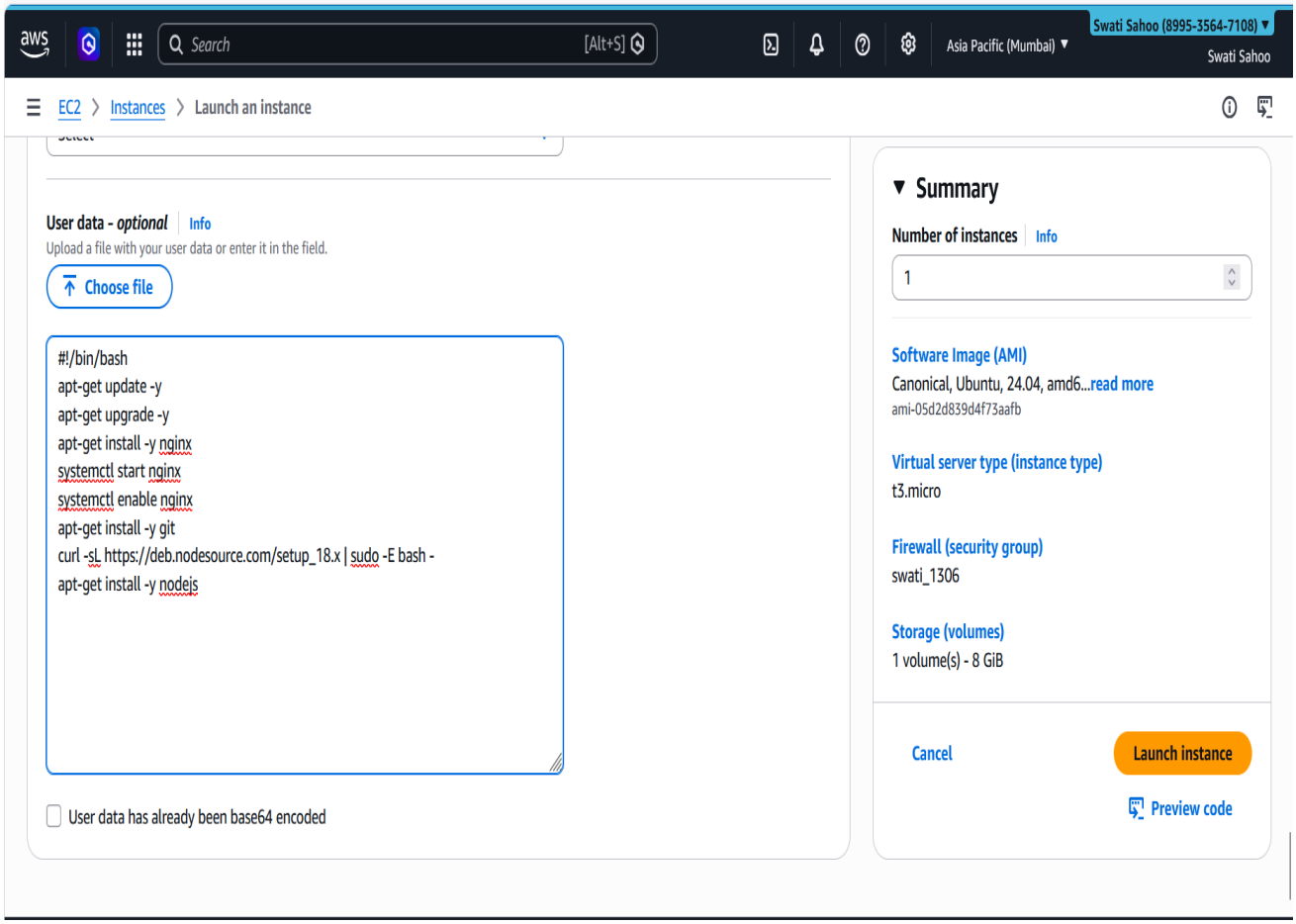
Cancel | Launch instance

Preview code

STEP 3: Add User Data Script

In Advanced Details → User Data, add:

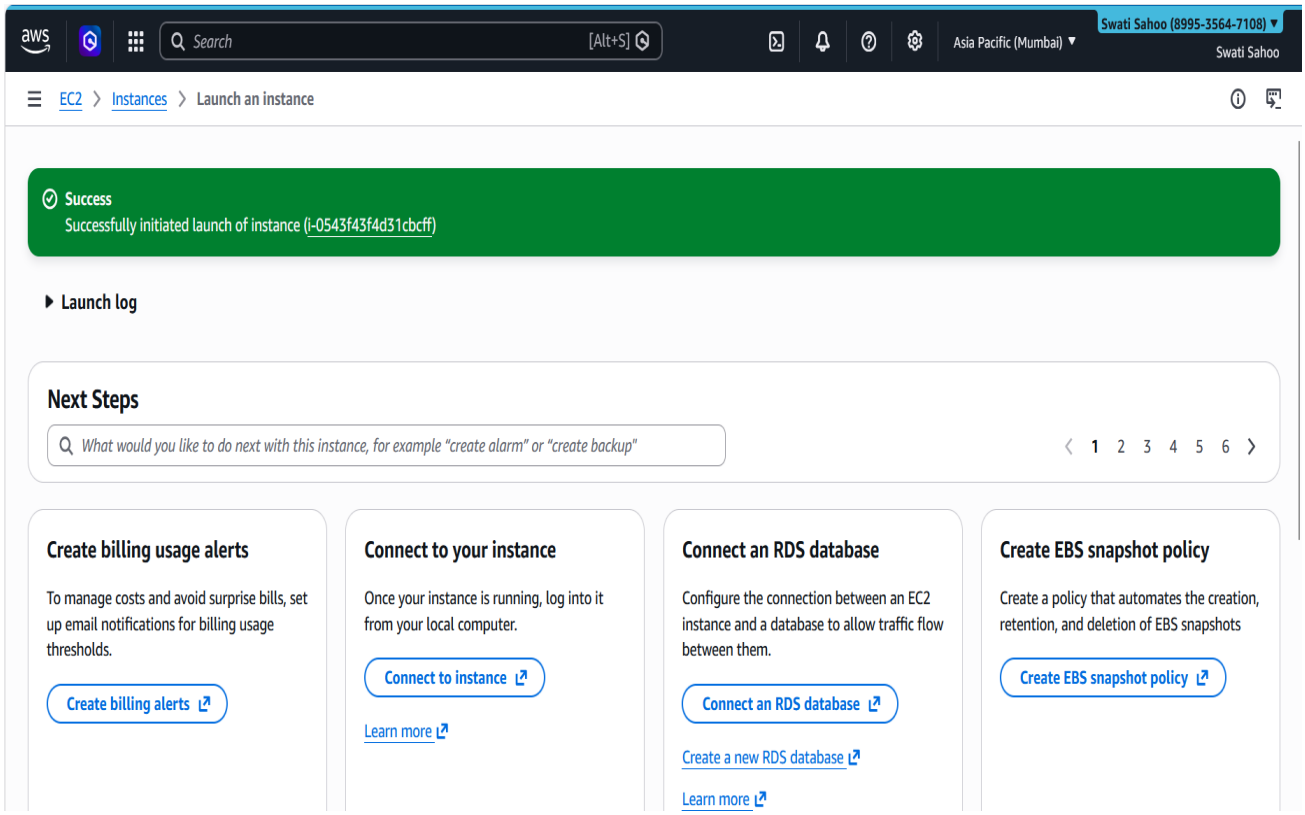
```
#!/bin/bash
apt-get update -y
apt-get upgrade -y
apt-get install -y nginx
systemctl start nginx
systemctl enable nginx
apt-get install -y git
curl -sL https://deb.nodesource.com/setup_18.x | sudo -E bash -
apt-get install -y nodejs
```



What this does:

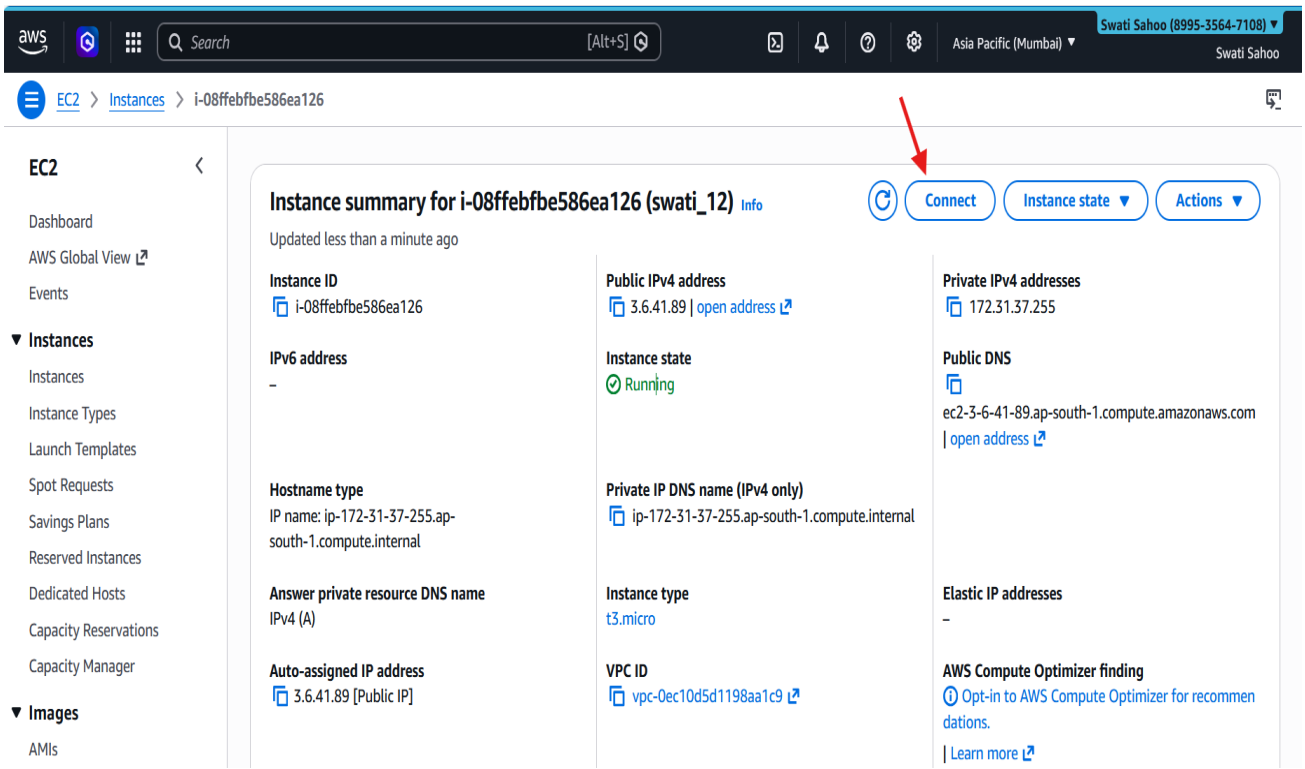
- Updates system
- Installs Nginx
- Installs Git
- Installs Node.js

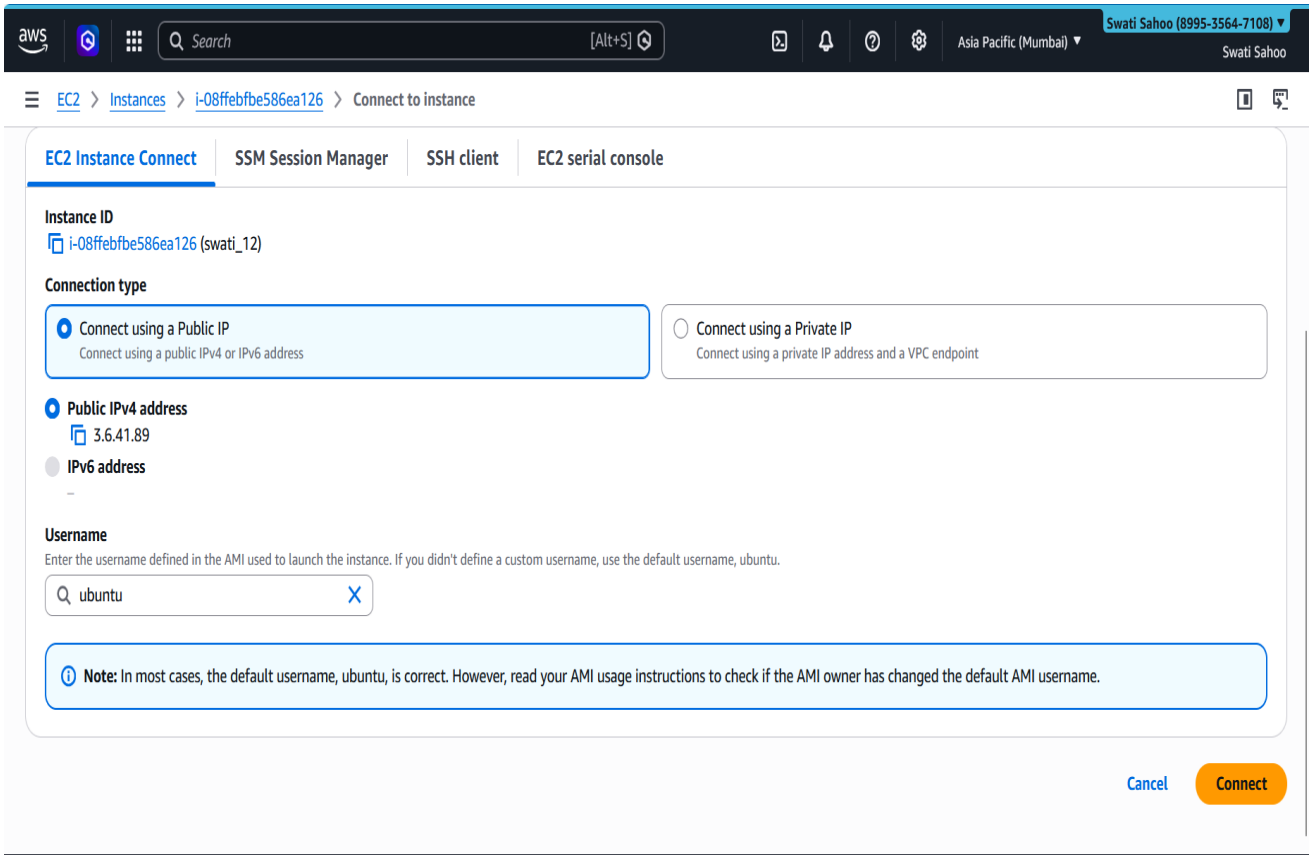
➢ Click **Launch Instance**



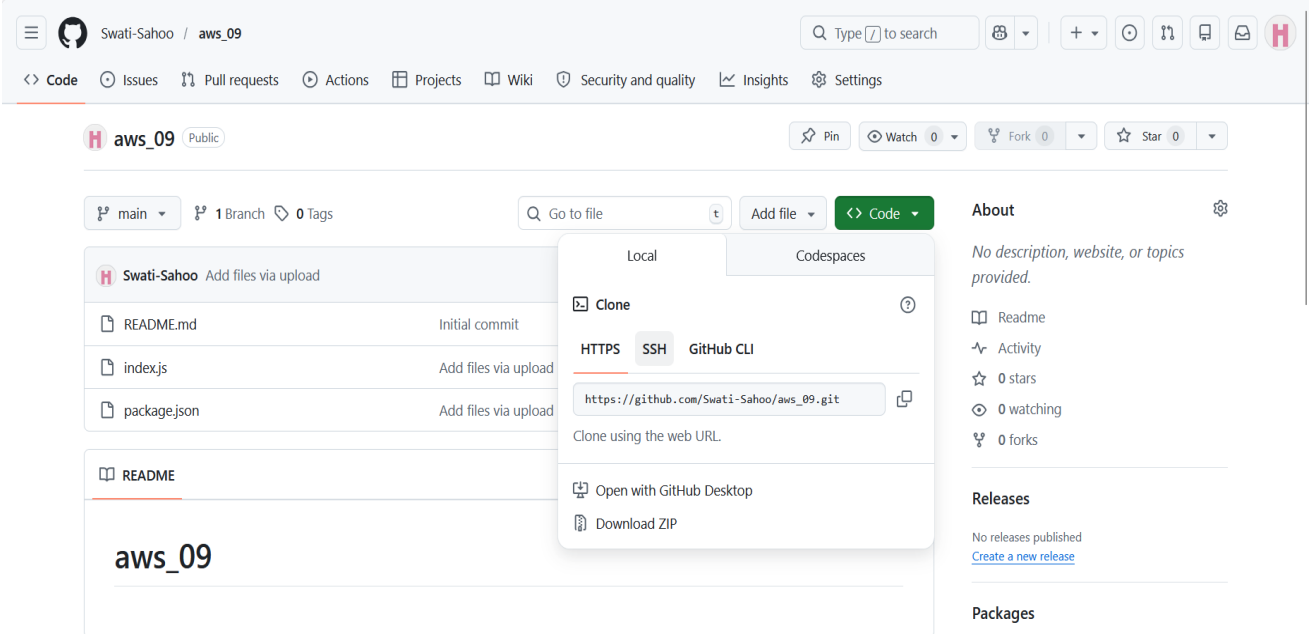
STEP 4: Connect to EC2

1. Go to Instances
2. Select your instance
3. Click **Connect**
4. Use **EC2 Instance Connect**
5. Username: ubuntu
6. Click **Connect**





->Terminal opens successfully.
->**Go to GITHUB Account** .
(ii)Open the which have node files , in my case 'aws_09' , and copy the REPO URL Link.



STEP 5: Clone GitHub Repository

Inside EC2 terminal:

git clone https://github.com/Swati-Sahoo/aws_09.git

cd aws_09

ls

```
The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-172-31-41-67:~$ git clonehttps://github.com/Swati-Sahoo/aws_09.git
git: 'clonehttps://github.com/Swati-Sahoo/aws_09.git' is not a git command. See 'git --help'.
ubuntu@ip-172-31-41-67:~$ git clone https://github.com/Swati-Sahoo/aws_09.git
Cloning into 'aws_09'...
remote: Enumerating objects: 7, done.
remote: Counting objects: 100% (7/7), done.
remote: Compressing objects: 100% (5/5), done.
remote: Total 7 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (7/7), done.
ubuntu@ip-172-31-41-67:~$ cd aws_09
ubuntu@ip-172-31-41-67:~/aws_09$ ls
README.md  index.js  package.json
ubuntu@ip-172-31-41-67:~/aws_09$
```

STEP 6: Install Dependencies

npm install

STEP 7: Run the Application

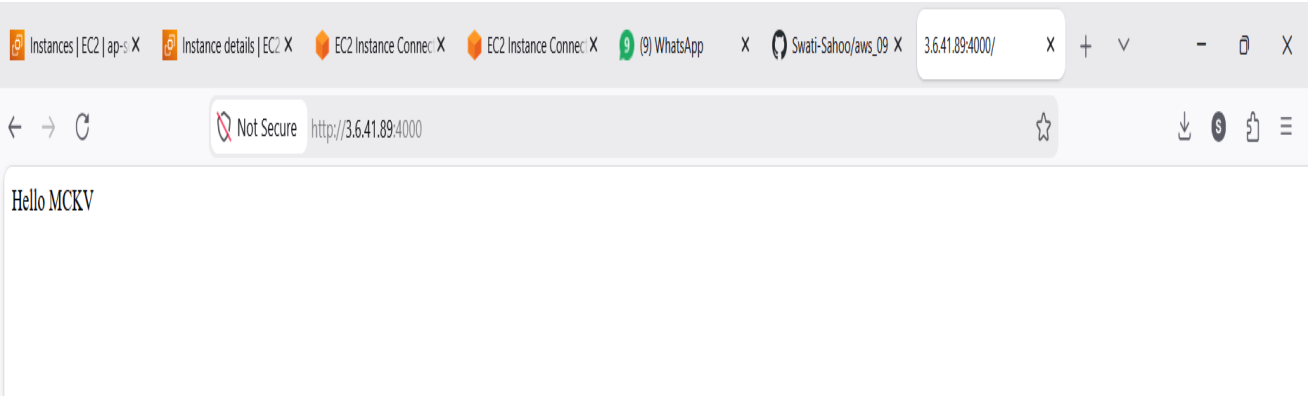
If your entry file is index.js:node index.js

```
remote: Total 7 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (7/7), done.
ubuntu@ip-172-31-41-67:~$ cd aws_09
ubuntu@ip-172-31-41-67:~/aws_09$ ls
README.md  index.js  package.json
ubuntu@ip-172-31-41-67:~/aws_09$ npm install

added 224 packages, and audited 225 packages in 28s

26 packages are looking for funding
  run `npm fund` for details

1 low severity vulnerability

Some issues need review, and may require choosing
a different dependency.

Run `npm audit` for details.
npm notice
npm notice New major version of npm available! 10.8.2 -> 11.12.1
npm notice Changelog: https://github.com/npm/cli/releases/tag/v11.12.1
npm notice To update run: npm install -g npm@11.12.1
npm notice
ubuntu@ip-172-31-41-67:~/aws_09$ node index.js
Started server
```

STEP 8: Access Application in Browser

1. Copy Public IPv4 address
   3.6.41.89
2. Open browser: http://3.6.41.89:4000/

➢ It loads → Deployment Successful



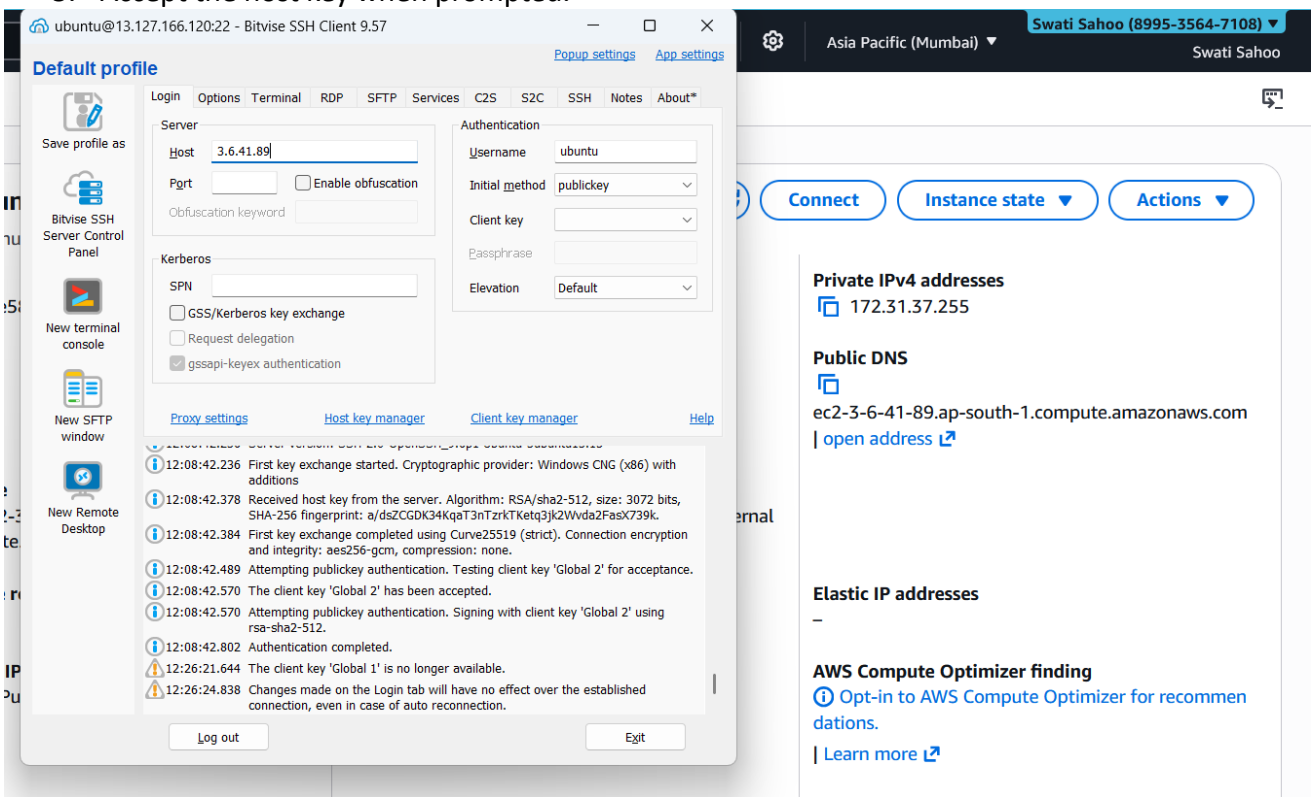
STEP 9: Connect to EC2 Using Bitvise SSH

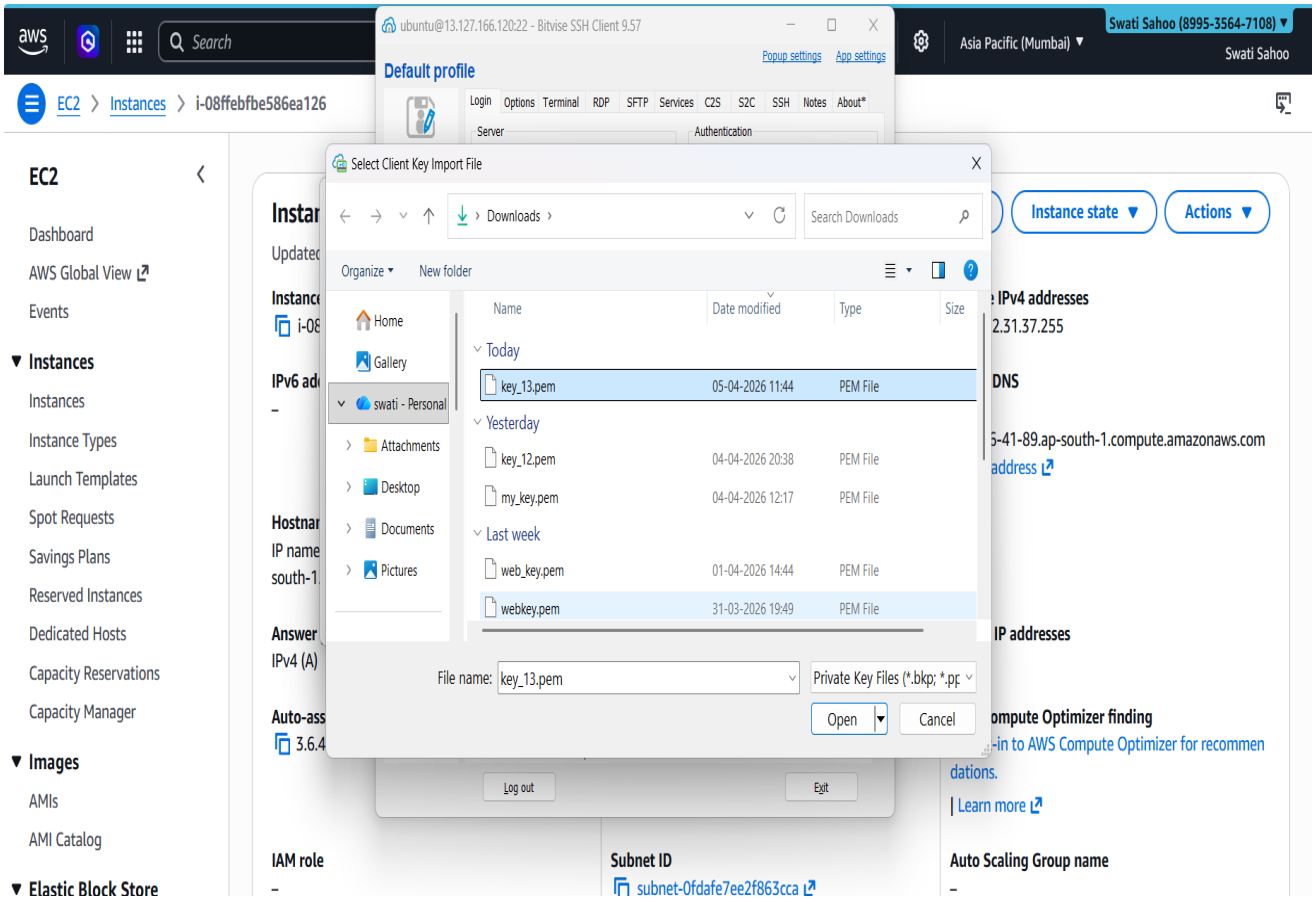
1. Open **Bitvise SSH Client**.
2. Enter:

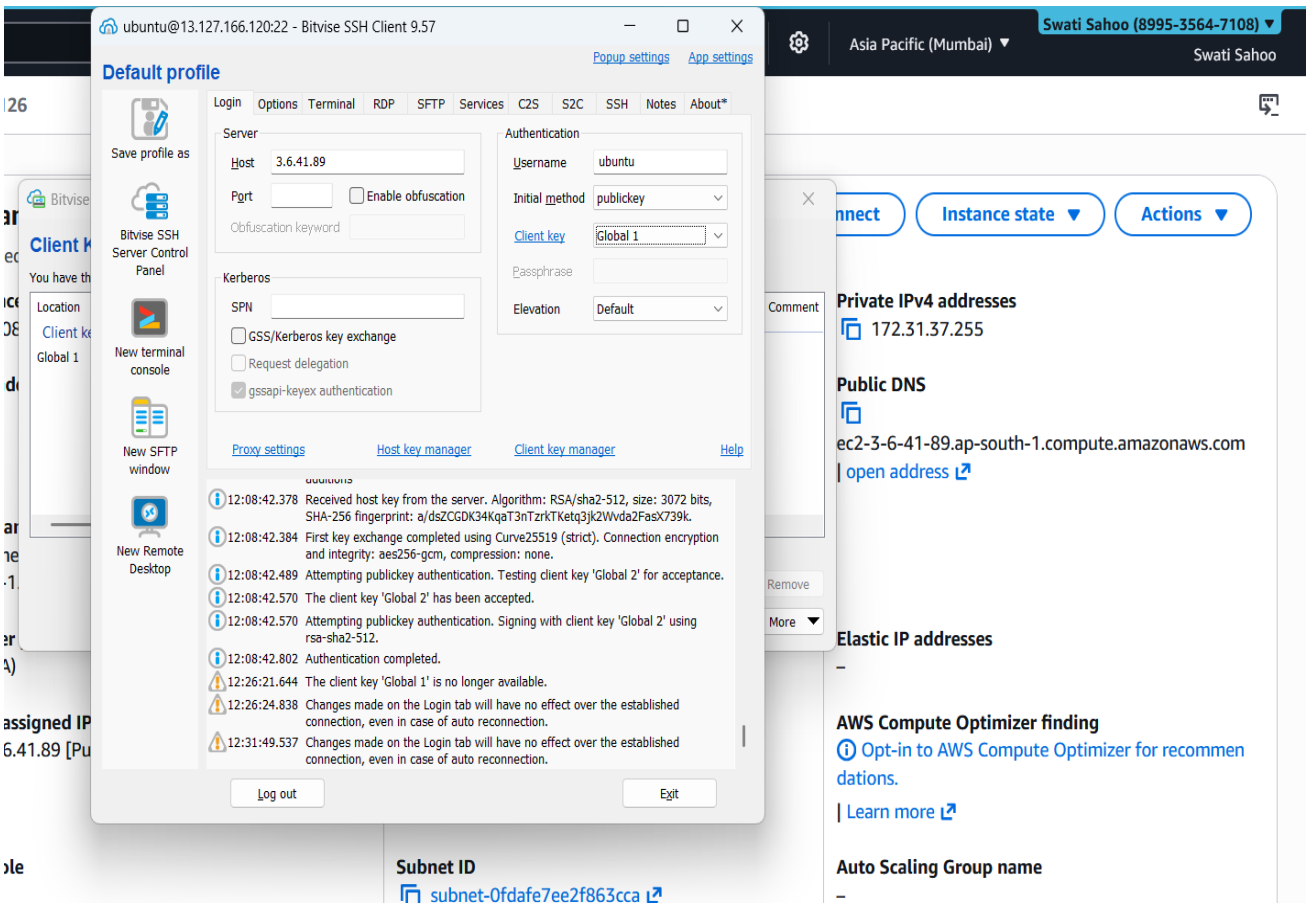
Host: 3.6.41.89

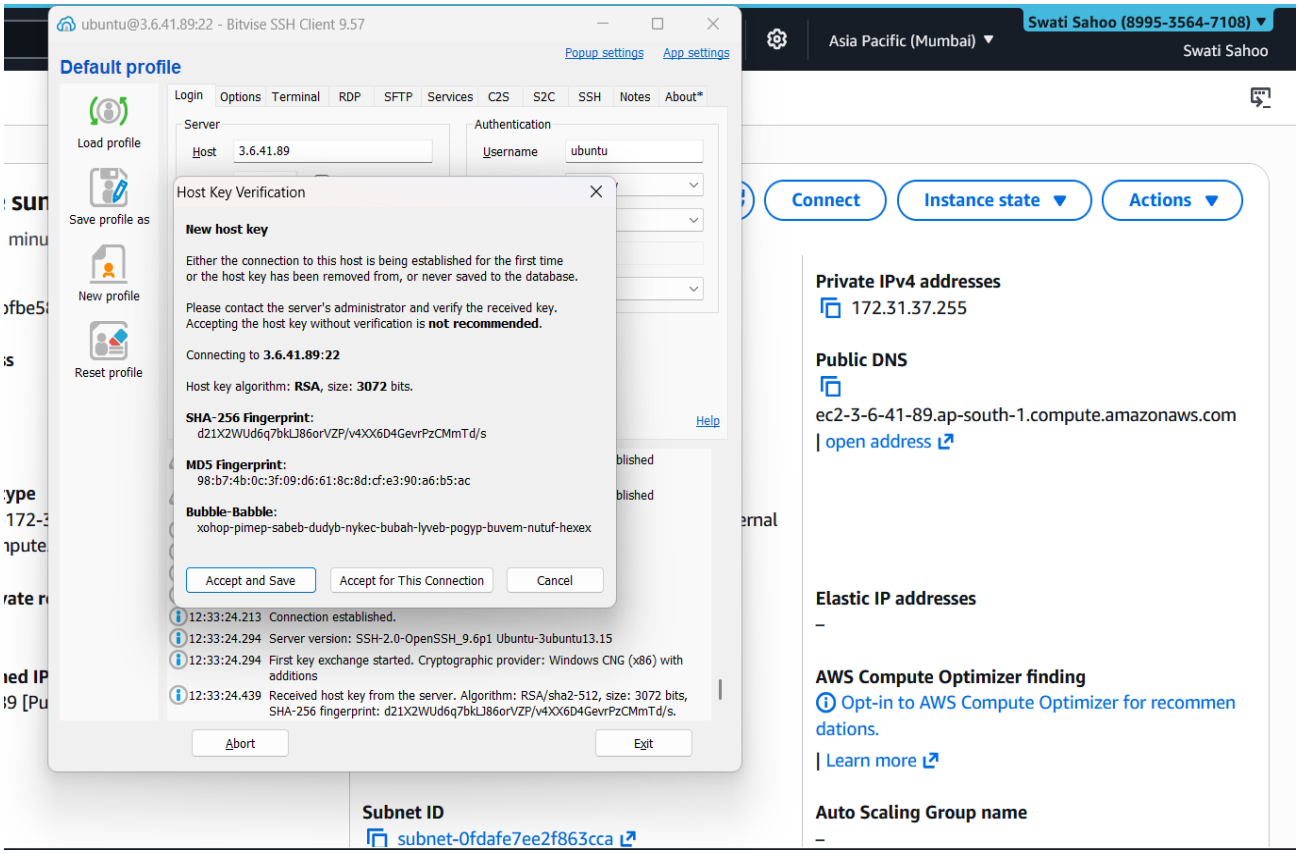
Username:ubuntu

3. Import the **key_13 key file**.
4. Click **Login**.
5. Accept the host key when prompted.




ubuntu@13.127.166.120:22 - Bitvise SSH Client 9.57
Default profile
Popup settings
App settings
Login Options Terminal RDP SFTP Services C2S S2C SSH Notes About*
Server
Authentication
Select Client Key Import File
Downloads
Search Downloads
Organize
New folder
Home
Gallery
swati - Personal
Attachments
Desktop
Documents
Pictures
Name
Date modified
Type
Size
Today
key_13.pem 05-04-2026 11:44 PEM File
Yesterday
key_12.pem 04-04-2026 20:38 PEM File
my_key.pem 04-04-2026 12:17 PEM File
Last week
web_key.pem 01-04-2026 14:44 PEM File
webkey.pem 31-03-2026 19:49 PEM File
File name: key_13.pem
Private Key Files (*.bkp; *.pr
Open
Cancel
Log out
Exit
Asia Pacific (Mumbai)
Swati Sahoo (8995-3564-7108)
Swati Sahoo
EC2 > Instances > i-08ffebfbe586ea126
EC2
Dashboard
AWS Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
Capacity Manager
Images
AMIs
AMI Catalog
Elastic Block Store
Instance state
Actions
IPv4 addresses
2.31.37.255
DNS
5-41-89.ap-south-1.compute.amazonaws.com
address
IP addresses
ompute Optimizer finding
-in to AWS Compute Optimizer for recommen
dations.
Learn more
IAM role
Subnet ID
subnet-0fdafe7ee2f863cca
Auto Scaling Group name


ubuntu@13.127.166.120:22 - Bitvise SSH Client 9.57
Default profile
Popup settings
App settings
Save profile as
Bitvise SSH Server Control Panel
New terminal console
New SFTP window
New Remote Desktop
Login Options Terminal RDP SFTP Services C2S S2C SSH Notes About*
Server
Host 3.6.41.89
Port
Enable obfuscation
Obfuscation keyword
Kerberos
SPN
GSS/Kerberos key exchange
Request delegation
gssapi-keyex authentication
Authentication
Username ubuntu
Initial method publickey
Client key Global 1
Passphrase
Elevation Default
Proxy settings
Host key manager
Client key manager
Help
12:08:42.378 Received host key from the server. Algorithm: RSA/sha2-512, size: 3072 bits, SHA-256 fingerprint: a/dsZCGDK34KqaT3nTzrkTKetq3jk2Wvda2FasX739k.
12:08:42.384 First key exchange completed using Curve25519 (strict). Connection encryption and integrity: aes256-gcm, compression: none.
12:08:42.489 Attempting publickey authentication. Testing client key 'Global 2' for acceptance.
12:08:42.570 The client key 'Global 2' has been accepted.
12:08:42.570 Attempting publickey authentication. Signing with client key 'Global 2' using rsa-sha2-512.
12:08:42.802 Authentication completed.
12:26:21.644 The client key 'Global 1' is no longer available.
12:26:24.838 Changes made on the Login tab will have no effect over the established connection, even in case of auto reconnection.
12:31:49.537 Changes made on the Login tab will have no effect over the established connection, even in case of auto reconnection.
Log out
Exit
Asia Pacific (Mumbai)
Swati Sahoo (8995-3564-7108)
Swati Sahoo
Instance state
Actions
Private IPv4 addresses
172.31.37.255
Public DNS
ec2-3-6-41-89.ap-south-1.compute.amazonaws.com
open address
Elastic IP addresses
AWS Compute Optimizer finding
Opt-in to AWS Compute Optimizer for recommen
dations.
Learn more
Subnet ID
subnet-0fdafe7ee2f863cca
Auto Scaling Group name



STEP 10: Navigate to Nginx Configuration Directory

After login, open **Terminal** inside Bitvise.

First move to root directory.

cd /

Check folders.

ls

Move to **etc directory**.

cd etc

Check files.

ls

Now go to **nginx folder**.

cd nginx

List files again.

ls

Go to **sites-available** directory.

cd sites-available

Check configuration files.

ls

You will see:

Default

```
ubuntu@ip-172-31-41-67:~$ cd /
ubuntu@ip-172-31-41-67:/$ ls
bin                boot  etc   lib                lib64       media  opt   root  sbin               snap  sys  usr
bin.usr-is-merged  dev   home  lib.usr-is-merged  lost+found  mnt    proc  run   sbin.usr-is-merged srv   tmp  var
ubuntu@ip-172-31-41-67:/$ cd etc
ubuntu@ip-172-31-41-67:/etc$ ls
ModemManager          cron.hourly          groff               ld.so.conf     modprobe.d             perl          screenrc          sysstat
PackageKit            cron.monthly         group               ld.so.conf.d   modules                pki           security          systemd
X11                   cron.weekly          group-              ldap           modules-load.d         plymouth      selinux           terminfo
acpi                  cron.yearly          grub.d              legal          mtab                   pm            sensors.d         timezone
adduser.conf          crontab              gshadow             libaudit.conf  multipath              polkit-1      sensors3.conf     tmpfiles.d
alternatives          cryptsetup-initramfs gshadow-            libblockdev    multipath.conf         pollinate     services          ubuntu-advantage
apparmor              crypttab             gss                 libibverbs.d   nanorc                 ppp           sgml              ucf.conf
apparmor.d            dbus-1               hdparm.conf         libnl-3        needrestart            profile       shadow            udev
apport                debconf.conf         hibagent-config.cfg locale.alias   netconfig              profile.d     shadow-           udisks2
apt                   debian_version       hibinit-config.cfg  locale.conf    netplan                protocols     shells            ufw
bash.bashrc           default              host.conf           locale.gen     network                python3       skel              update-manager
bash_completion       deluser.conf         hostname            localtime      networkd-dispatcher    python3.12    sos               update-motd.d
bash_completion.d     depmod.d             hosts               logcheck       networks               rc0.d         ssh               update-notifier
bindresvport.blacklist dhcp                hosts.allow         login.defs     newt                   rc1.d         ssl               usb_modeswitch.conf
binfmt.d              dhcpcd.conf          hosts.deny          logrotate.conf nftables.conf          rc2.d         subgid            usb_modeswitch.d
byobu                 dpkg                 init.d              logrotate.d    nginx                  rc3.d         subgid-           vconsole.conf
ca-certificates       e2scrub.conf         initramfs-tools     lsb-release    nsswitch.conf          rc4.d         subuid            vim
ca-certificates.conf  ec2_version          inputrc             lvm            opt                    rc5.d         subuid-           vmware-tools
chrony                environment          iproute2            machine-id     os-release             rc6.d         sudo.conf         vtrgb
cloud                 ethertypes           iscsi               magic          overlayroot.conf       rcS.d         sudo_logsrvd.conf wgetrc
console-setup         fstab                issue               magic.mime     overlayroot.local.conf resolv.conf   sudoers           xattr.conf
credstore             fuse.conf            issue.net           manpath.config pam.conf               rmt           sudoers.d         xdg
credstore.encrypted   fwupd                kernel              mdadm          pam.d                  rpc           supercat          xml
cron.d                gai.conf             landscape           mime.types     passwd                 rsyslog.conf  sysctl.conf       zsh_command_not_found
cron.daily            gnutls               ld.so.cache         mke2fs.conf    passwd-                rsyslog.d     sysctl.d
ubuntu@ip-172-31-41-67:/etc$
```

```
ubuntu@ip-172-31-37-255:/etc$ cd nginx
ubuntu@ip-172-31-37-255:/etc/nginx$ ls
conf.d       fastcgi_params  koi-win    modules-available  nginx.conf    scgi_params     sites-enabled  uwsgi_params
fastcgi.conf koi-utf         mime.types modules-enabled    proxy_params  sites-available snippets       win-utf
ubuntu@ip-172-31-37-255:/etc/nginx$ cd sites-available
ubuntu@ip-172-31-37-255:/etc/nginx/sites-available$ ls
default
ubuntu@ip-172-31-37-255:/etc/nginx/sites-available$
```

STEP 11: Edit Nginx Default Configuration

Open the configuration file.

sudo nano default

Inside the file find:

location /

Modify it as follows:

```
location / {
    proxy_pass http://localhost:4000;
    proxy_http_version 1.1;
    proxy_set_header Upgrade $http_upgrade;
    proxy_set_header Connection 'Upgrade';
    proxy_set_header Host $host;
    proxy_cache_bypass $http_upgrade;
}
```

This configuration tells **Nginx to forward all traffic from port 80 to port 4000** where my Node.js app is running.

```
        root /var/www/html;

        # Add index.php to the list if you are using PHP
        index index.html index.htm index.nginx-debian.html;

        server_name _;

        location / {
              proxy_pass http://localhost:4000;
    proxy_http_version 1.1;
    proxy_set_header Upgrade $http_upgrade;
    proxy_set_header Connection 'Upgrade';
    proxy_set_header Host $host;
    proxy_cache_bypass $http_upgrade;

        }
```

Save and exit:

Press:Ctrl + X

Then press:Y

Press:Enter

STEP 12: Restart Nginx Server

Now go back to main directory.

cd ..
cd ..

Restart Nginx service.

sudo systemctl restart nginx

This reloads the new configuration.

```
ubuntu@ip-172-31-37-255:/etc/nginx/sites-available$ sudo nano default
ubuntu@ip-172-31-37-255:/etc/nginx/sites-available$ cd ..
ubuntu@ip-172-31-37-255:/etc/nginx$ cd ..
ubuntu@ip-172-31-37-255:/etc$ sudo systemctl restart nginx
ubuntu@ip-172-31-37-255:/etc$
```

STEP 13: Access Application Without Port

Open **Browser (Incognito mode)**.

Paste only the **Public IPv4 address**:

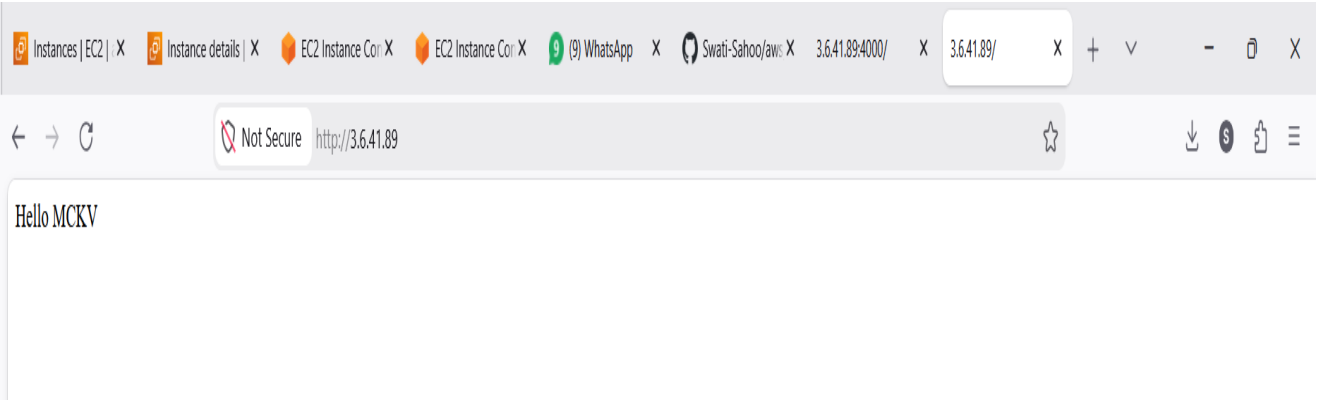
http:// 3.6.41.89

Now the application will load **without :4000**.

(Previously: http:// 3.6.41.89:4000)

Now:

http:// 3.6.41.89



Deployment successful.

Conclusion :

The project was successfully deployed on an AWS EC2 instance. Initially the application was accessed using port 4000. Nginx was configured as a reverse proxy to redirect incoming HTTP requests from port 80 to port 4000. After restarting the Nginx service, the application could be accessed using only the public IPv4 address without specifying the port number, confirming successful deployment.